> **“A hospital… should do the sick no harm.”**
>
> FLORENCE NIGHTINGALE, BRITISH NURSE AND PHILANTHROPIST, 1859

the role of the liver and pancreas in the digestive system and the function of the kidneys. This in turn gave them new ways of diagnosing illnesses, such as identifying kidney disease by analyzing the chemicals found in a patient’s urine.

Surgery

Once doctors began to think of disease affecting the solid organs of the body, they realized they might be able to cure patients by cutting out the diseased part. However, in the 18th century, surgical operations were used only as a last resort, because of the pain involved in cutting into the patient’s body, and the high probability of wound infection.

From the beginning of the 19th century, surgery began to change with the introduction of anesthetics—the British surgeon Robert Liston’s use of ether during an amputation had an impact throughout Europe. Now surgeons could work more slowly, reducing blood loss by tying up blood vessels, and carefully cutting out damaged or diseased tissue. Around the same time, they tackled the problem of wound infection: first by keeping their instruments and the wound clean, and later by using antiseptics to kill the germs that caused infection. At the end of the 19th century, modern aseptic techniques were introduced to eliminate contamination from the operating room: staff wore gowns, masks, and rubber gloves, and every instrument was sterilized. This allowed surgeons to operate safely, and to devise a range of operations to cure diseases.

Therapy

Although the practice of bloodletting—draining a patient of blood in the hope it would cure their illness—was still used in the 19th century, by 1900 doctors had abandoned most of the old “humoral” therapies (see BEFORE). However, there were still few effective remedies. Opium was used to relieve pain, but as an addictive substance it was not without risk. Doctors could treat serious injuries and broken limbs with the aid of anesthetics and improved surgical methods. Quinine (made from the bark of the cinchona tree) was used to treat malaria, and digitalis (made from foxgloves) was an effective treatment for heart conditions. In 1897, aspirin (based on a chemical found in willow bark) was launched. Salvarsan, the first synthetic drug, was introduced in 1910, and was prescribed in cases of syphilis. However, many remedies did not cure disease, but only soothed the symptoms—for example, cough syrup eased coughing, and poultices reduced swellings. Bed rest, nutritious food, and careful nursing were still the basis of medical care for most forms of illness.

INVENTIONS

SCOPES AND X-RAYS

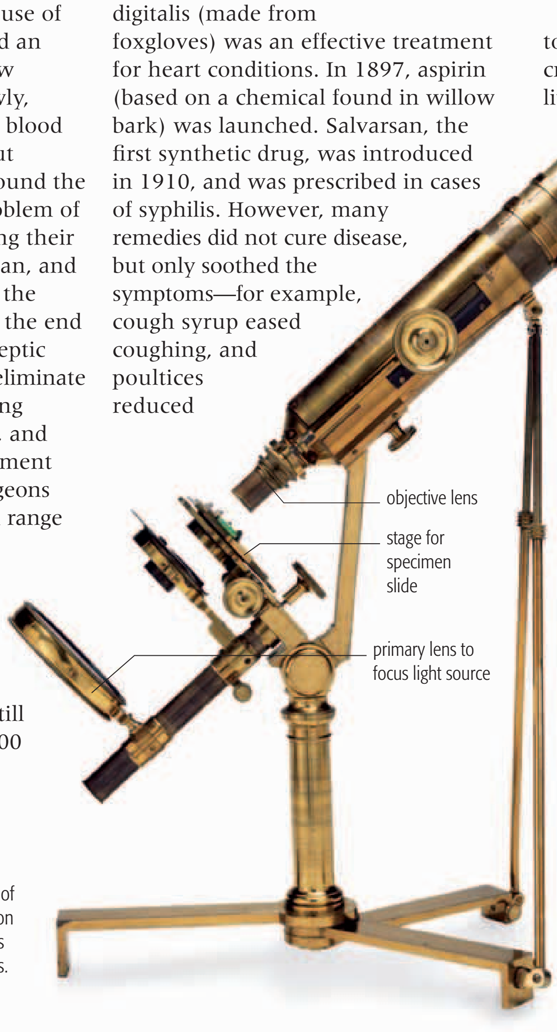
The idea that disease was located in the body’s organs led to a string of inventions helping doctors to “see” inside of living patients. In 1816 René Laennec created the first crude stethoscope when he rolled up some papers into a tube, and discovered he could hear sounds made by the lungs and heart. The 19th century saw the invention of a number of “scopes” for seeing into the ear, eye, and the throat. The ability to see inside the living body became a reality in 1895 when Wilhelm Röntgen discovered that X-rays passed through flesh but not bone. X-ray images, such as this one from 1895, became an invaluable tool in helping doctors to repair damaged limbs.

30 SECONDS was said to be all it took for the 19th-century surgeon Robert Liston to amputate a human leg—without administering an anesthetic.

Caring for the sick

Hospitals were crucial to the development of new medical ideas and practices, and in turn these medical innovations completely transformed the nature of the hospital. In the medieval period, hospitals across Europe were originally linked to monasteries and abbeys and were created to offer food and shelter—literally, hospitality (see pp.200–01). They cared for the poor, the sick, and religious pilgrims. In the 18th century, hospitals were still charities caring for the sick and poor. In the 19th century, they became centers for treatment, open to all who could afford to pay. Operating rooms were built to allow students to view surgery, with specialized lights and equipment, and lined with glass and tiles that could be easily cleaned. Some teaching hospitals invested in new equipment, such as the first X-ray machines (see above). They also began to offer specialty treatments for complaints of the ear and eye, or conditions affecting children. As a result, hospitals treated a greater number of seriously ill patients, and the amount of staff, especially nurses, grew enormously to provide round-the-clock care. Nurses, who were almost always women, played a crucial role in the new hospitals. They kept the wards scrupulously clean and neat to limit the possibility of infection, monitored patients’ symptoms, and administered medicines. However, even experienced and skilled nurses were expected to follow the doctors’ directions at all times.

While the relatively well-equipped hospitals were able to treat a wider range of conditions, many people, particularly the poor, continued to die of infectious diseases transmitted by bacteria and viruses. Cholera, typhoid fever, tuberculosis, and influenza killed thousands. Children were particularly vulnerable, and many died from measles, mumps, diphtheria, and scarlet fever. Doctors could do little to cure these diseases, but they could control their spread. Increased supplies of clean water and improved sanitation helped to clean up cities (see pp.322–25) and stop the spread of waterborne infections. Patients suffering from infectious diseases were isolated in special hospitals, or in their homes. Quarantine was used, so those traveling from areas where disease was present were forced to stay away from towns until it was certain that they were not carrying an infection.



Microscope
From 1830, the microscope played a crucial role in understanding the structure of the body. Doctors diagnosed cancer by examining tissues under the microscope, and confirmed infection by identifying disease-causing bacteria.

Early operating theatre
This 1898 photograph of an operating room reveals that the idea of asepsis (the reduction of contamination) spread very slowly. The surgeon wears an apron, but no mask, and the room is filled with observers wearing ordinary clothes.

AFTER

In the late 19th and 20th centuries, the greatest medical developments came in the form of new treatments.

VACCINES
Smallpox vaccination had been developed in the 1790s, based on the **chance observation** that milkmaids who caught cowpox (a mild disease) did not catch smallpox. Other forms of immunization were developed as doctors began to understand how the body resisted disease. In the 1880s, French scientist Louis Pasteur developed a vaccine against rabies based on a **weakened form of the virus**. This was followed by vaccines against tetanus and diphtheria in the 1890s. **Mass immunization** against common childhood diseases came in the 20th century—with a vaccine against polio in 1957 and against measles and mumps in the 1960s.

LOUIS PASTEUR

NEW DRUGS
The first drugs that treated infections were produced in the 1930s, but they were replaced by the **discovery of penicillin**, which became available in the 1940s **464–65 ››**. Before then, most drugs were derived from plants, but since the 1950s, **synthetic drugs** have predominated.